TCS — Aptitude Test 4

Questions 76–100 of 100 · 25 questions

Question 76. How many 6 digit numbers can be formed using digits 1,2,3,4,5,6 without repetition such that hundreds digit is greater than ten's digit and ten's is greater than ones digit?

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Solution:
Number of ways of selecting 3 digits out of 6 digits for the last three digits
= 6C3 = 20.
3 numbers can be arranged in 6 ways out of which only one arrangement will have hundreds place greater than tens place and tens place greater than ones place.
So, in all these selections there is only one arrangement in each which follows the condition.
Number of arrangements for the last three digits = 20 x 1 = 20
The remaining three digits can be arranged in 3! Ways = 6 ways.
Total number of arrangements for the above condition = 6 x 20 = 120 ways.

Question 77. How many 9 digit numbers can be formed using the digits 1, 2, 3, 4, 5 which are divisible by 4 if the repetition is allowed?

Solution:
To be divisible by 4, the last two digits of the number must be divisible by 4.
The number of two digit numbers which are divisible by 4 that can be formed using 1,2,3,4,5 are
12, 24, 32, 44 and 52.
We have 5 different cases.

Question 78. How many positive numbers not greater than 4300 can be formed using the digits 0,1,2,3,4 where repetition is allowed?

Solution:
All 1 digit, 2 digit and 3 digit numbers are less than 4300.
Number of 1 digit numbers that can be formed from 0,1,2,3,4 = 4 (zero is neither positive nor negative.
Number of two digit numbers
= 4 x 5 = 20
Number of three digit numbers
= 4 x 5 x 5 = 100
Number of 4 digit numbers starting with 1 or 2 or 3
= 3 x 5 x 5 x 5 = 375
Number of four digit numbers starting with 40 or 41 or 42
= 3 x 5 x 5 = 75.
Total numbers less than 4300
= 20 + 100 + 375 + 75
= 570

Question 79. Find the rank of the word GOOGLE if all the words which can be formed by permuting letters of this word without repetition are arranged in dictionary in alphabetical order?

Solution:
Arranging the letters in alphabetical order, we get
EEGGLE.
Assign numbers in ascending order, we get, E = 1; G = 2; L = 3 and O = 4
The first number will be,
122344
The number arrangement according to the question = 244231
Total number of arrangements for the word GOOGLE = 6!/2!x2! = 180.

If the first number is 1, the remaining 5 digits can be arranged in 5!/(2!x2!) = 30 ways.
Since the starting number of the given word is 2, the rank is definitely greater than 30.
If the first number is 2, the remaining 5 digits can be arranged in 5!/(2!) = 60 ways.
The rank of the given word is between 30 and 90.
If the second number is 1, the remaining 4 digits can be arranged in 4!/2! = 12 ways.
So the rank is above 42.
If the second number is 2, the remaining digits can be arranged in 4!/2! = 12 ways.
So the rank is above 54.

If the second number is 3, the remaining digits can be arranged in 4!/2! = 12 ways.
So the rank is above 66.
if the second number is 4, the remaining digits can be arranged in 4! = 24 ways.
The rank is between 66 and 90.
If the third number is 1, the remaining 3 digits will be arranged in 3! = 6 ways.
If the third number is 2, the remaining digits will be arranged in 3! = 6ways
If the third digit is 3, the remaining 3 digits will be arranged in 3! = 6 ways.
If the third number is 4, the remaining digits will be arranged in 3! = 6 ways.
The rank is between 84 and 90.
If the fourth number is 1, the remaining 2 digits can be arranged in 2! = 2 ways.
If the fourth number is 2, the remaining 2 digits will be arranged in 2! = 2 ways
The rank is between 87 or 88.
If the fifth number is 1 the rank is 87.
Since the fifth number is 3, the rank is 88.

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Question 80. In how many ways can the digit of the number 2233558888 be arranged so that the odd digits are placed in the even positions?

Solution:

The odd numbers in 2233558888 are 3,3,5,5.
The blank spaces below are the even places.

Question 81. 7 noun, 5 verbs and 2 adjectives are written on blackboard. We can form a sentence by choosing 1 from each available set without caring it makes sense or not. What is the number of ways of doing this?

Question 82. A teacher sets a question paper consisting of 8 question of total 40 marks such that each question contains

**minimum
two
marks
and
integer
number of marks. In how may ways this can be done?**

Question 83. There are 4 boxes colored red, yellow, green and blue. If 2 boxes are selected, how many combinations are there for at least one green box or one red box to be selected ?

E_E_E_E_E_
Out of 5 even places four should be selected for the odd numbers.
It can be done in 5C4 = 5 ways.
In each selection the odd numbers can be arranged in 4!/(2!x2!) = 6 ways.
Total ways for odd numbers = 6x5 = 30
The remaining 6 places can be arranged in 6!/(2!x2!x2!) = 90 ways.
Total number of arrangements = 30 x 90 = 2700 ways.

Solution:
The number of ways of selecting 2 boxes out of 4 boxes = 4C2 = 6.
Out of these 6 selections, one selection will have yellow and blue (neither red nor green)
The remaining 5 selections will have at least one red or one green.
Answer = 5

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Question 84. In how many ways can 7 different objects be divided among three persons so that either one or two of them do not get any object?

Solution:
If two of them do not get any object, it can be done in 3 ways. Either first or second or the third person will get all the objects.
If one of them did not get any object, the remaining two
persons can have the
number
of
objects in
the following

manner (1,6) or (2,5) or (3, 4).
The two persons out of the three can be selected in 3C2 = 3 ways.
And in each selection the number of objects will interchange.
Total ways when two persons are having the objects = 3 x 2 =6
Total ways = 3 + 6 = 9 ways.

Solution:
Number of ways of selecting 1 noun and 1 verb and 1 adjective is
7C1 x 5C1 x 2C1 = 7 x 5 x 2 = 70
The number of ways in which the three words can be arranged = 3! = 6
Total number of sentences
= 6 x 70 = 420.

Question 85. In how many ways we can distribute 10 identical looking pencils to 4 students so that each student get at least one pencil?

Solution:
The minimum marks for all the 8 questions = 8 x 2 = 16
Remaining marks = 40 – 16 = 24.
By applying non negative integer formula, we get
(24+8-1)C(8-1)
= 31C7
= 2629575
The questions can be taken in 2629575 ways.

Solution:
Give each student one pencil. Remaining number of pencils = 6
Using negative integer formula, we get
(6+4-1)C(4-1) = 9C3 ways
The number of ways
= 84 ways.

Question 86. It is dark in my bedroom and I want to get two socks of the same color from my drawer, which contains 24 red and 24 blue socks. How many socks do I have to take from the drawer to get at least

two socks of the same color?

Solution:
There are two different colors of socks.
By picking the first socks if it is red, the second socks may be red or blue. In thee second draw it is not guaranteed to get two socks of same color.
In the third draw you will get either red or blue. In this draw it is sure to have either the second red sock or the second blue sock.
The number of draws required = 3.

Question 87. In how many ways a team of 11 must be selected from 5 men and 11 women such that the team must comprise of not more than 3 men?

Solution:
The number of men should not be more than 3.
The different cases are:
0 men and 11 women or
1 man and 10 women or
2 men and 9 women or
3 men and 8 women.
The total ways of selecting
=(5C0x11C11) + (5C1x11C10) + (5C2x11C9) + (5C3+11C8)
= (1x1) + (5x11) + (10x55)+ (10x165)
= 1 + 55 + 550 + 1650
= 2256 ways.

Question 88. At the beginning of a peace conference between two tribes, each person shook hands precisely once with every other member of his own tribe. A total of 3856 handshakes(each pair meeting counting as one handshake). After the conference each person shook hands precisely once with every person in the other tribe. A total of 3525 handshakes. How many people attended

the conference?

Solution:
Assume that the number of persons in first tribe is ‘a’ and the second tribe is ‘b’.
Number of handshakes between the members of first tribe = aC2
Number of handshakes between the members of second tribe = bC2
aC2 + bC2 = 3856
If there are 3 persons in one group and 4 persons in another group, each person in first group will have four handshakes with the persons in the other group.
Total handshakes will be 3 x 4 = 12.
From the question, when both the groups shook hand with each other, there were 3525 handshakes.
So, a x b = 3525
Factors of 3525 = 5 x 5 x 3 x 47
One of the tribe has 5x5x3 = 75 and
The other tribe has 47 members.
Total members = 75 + 47 = 122

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Question 89. A lady took out jacket and gloves, which are available in blue 16, yellow 40 and red 36. Power goes off, she can distinguish between gloves and jacket but not in colors. What's the possibility that she will pick up pair of gloves from any one color?

Solution:
Ways of selecting 2 blue gloves from 16
= 16C2 = 120
Ways of selecting 2 yellow gloves
= 40C2 = 780
Ways of selecting2 red gloves
= 36C2 = 630
Ways of selecting 2 gloves
= 92C2 = 4186
Probability
= (120 + 780 + 630)/4186
= 1530/4186

Question 90. What is probability that 5 letters can not put in to their right envelope?

Solution:
Let us name the envelops and letters
a, b, c, d, e are envelops
1, 2, 3, 4, 5 are letters.

Envelop ‘a’ must not have letter 1. So it can be arranged in 4 ways.
Envelop ‘b’ must not have letter 2. So it can be arranged in 3 ways.
Envelop ‘c’ must not have letter 3. So it can be arranged in 2 ways.
Envelop ‘d’ must not have letter 4. So it can be arranged in 1 way.
If the first 4 envelops does not have corresponding letters, the fifth envelop will also have the wrong letter.
Total ways of arranging in wrong manner = 4x3x2x1 = 24
Total ways of arranging 5 letters I 5 envelops = 5! = 120
Probability = 24/120 = 1/5

Question 91. 3 houses are available in a locality. 3 persons apply for the houses. Each apply for one house without consulting others. The probability that all 3 apply for the same house is:

Solution:
Number of ways of selecting one house by one person = 3C1
Number of ways of selecting one house by three persons = 3C1 x 3C1 x 3C1= 27
Number of ways in which all the three persons select House number 1 or 2 or 3 = 3
Probability = 3/27 = 1/9

Solution:
To get the maximum possible amount, we must have four 5Rs. coin.
Ways of selecting four 5Rs coins = 5C4 = 5
Ways of selecting 4 coins out of (5+7+8= 20) 20 coins = 20C4 = 4845
Probability = 5/4845
= 1/969

Question 92. Thirty days are in September, April, June and November. Some months are of thirty one days. A month is chosen at random. Then its probability of having exactly three days less than maximum of 31 is?

Question 93. A car manufacturer produces only red and blue models which come out of the final testing area at random. What are the odds that five consecutive cars of same color will come through the test area at any one time?

Solution:
Probability of all five cars to be red
= (½)(½)(½)(½)(½) = (1/32)
Probability of all five cars being blue
= (½)(½)(½)(½)(½) = (1/32)
Probability of all five cars being red or blue
= 1/32 + 1/32
= 1/16

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Question 94. A bag contains 5 five-rupee coins, 8 two-rupee coins and 7 one-rupee coins. If four coins are drawn from the bag at random, then find the odds in favor of the draw yielding the maximum possible amount.

Question 95. Four people each roll a die once. Find the probability that at least two people will roll the same number?

Solution:
Total number of results when four dice are rolled = 64 = 1296
Number of ways in which at least two people roll same number = total results – number of ways in which all results are different.
Number of ways in which all results are different = 6 x 5 x 4 x 3 = 360
Probability = 1 – (360/1296)
= 1 – 5/18
= 13/18

Solution:
Three days less than 31 days = 28 days.
The month of February has 28 days.
February also has 29 days in a leap year.

Out of every 4 years, three years have 28 days in February and one year has 29 days in February.
Number of months in four years = 48
Number of months with 28 days = 3
Probability = 3/48

Question 96. One card is lost out of 52 cards. Two cards are drawn randomly. They are spade. What is the probability that the lost card is also spade?

Solution:
Assume that initially 2 cards are taken out and the two cards are spade.
Now there will be 11 spades out of 50 cards.
The probability of losing one spade = 11/50

Question 97. A bag contains 3 black and 3 white boxes, second bag contains 5 black and 1 white boxes, and finally third bag contains 4 black and 5 white boxes. If a box is chosen at random from one of these bags, the probability that it is not a white box is ?

Solution:
Probability of not selecting a white box = probability of selecting black box.
Total ways of selecting one box
= ways of selecting one box from bag 1
+ ways of selecting one box from bag 2
+ ways of selecting one box from bag 3
= 6C1 + 6C1 + 9C1 = 21
Ways of selecting one black box from bag 1, 2 and 3 = 3C1 + 5C1 + 4C1 = 12
Probability = 12/21 = 4/7

Question 98. A basket contains 6 red balls, 5 blue balls, 4 green balls and 3 white balls. Five balls are to be drawn together at random, then what is the probability that there is at least a ball of each color?

Solution:
Total ways of selecting 5 balls = 18C5
If there must be 1 ball in each color,

there should be only one color with two balls out of the five.
Ways of selecting 2R, 1B, 1G, 1W
= 6C2 x 5C1 x 4C1 x 3C1 = 900
Ways of selecting 1R, 2B, 1G, 1W
= 6C1 x 5C2 x 4C1 x 3C1 = 720
Ways of selecting 1R, 1B, 2G, 1W
= 6C1 x 5C1 x 4C2 x 3C1 = 540

Ways of selecting 1R, 1B, 1G, 2W
= 6C1 x 5C1 x 4C1 x 3C2 = 360
Total ways of selecting 5 balls according to the condition
= 900 + 720 + 540 + 360
= 2520
Probability = 2520/(18C5)
= 2520/8568

Question 99. Three dice are rolled. What is the probability of getting the sum as 13?

Solution:
Total number of results when three dice are rolled = 63 = 216
Three numbers add up to 13. The possibilities are:
(3,4,6), (4,4,5), (5,5,3), (6,5,2), (6,6,1)
In each case the results can be arranged in 3! ways.
Total number of results which give sum 13 = 5 x 3! = 30
Probability = 30/216
= 5/36

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Question 100. If f(x) = 2x+2 what is f(f(3))?

Solution:
f(x) = 2x + 2
f(3) = 2(3) + 2
f(3) = 8
f(f(3)) = f(8)
= 2(8) + 2
= 18

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